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Serotonin in the Mitochondria Means Your Cells Are Fighting to Survive

Installment three in our series on understanding the truth about SSRIs.



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The next installment in [our series](#) on the true nature of SSRIs gets into the true nature of serotonin and its powerful impact on energy metabolism.

We will see that serotonin comes to rescue the mitochondria when they are fighting for survival.

Unlike melatonin, it does not hang out, wait, and keep watch in the mitochondria. It comes like a firefighter to defend the mitochondria under rare conditions of extreme stress.

Nevertheless, if you view yourself as a population of cells, you will realize that “rare” emergencies are happening *all the time*.

At any given moment there is *always* a fire to put out.

You may, at the organismal level, be fine. You could be thriving. But somewhere you have a mitochondrion under stress and serotonin is helping put the fire out.

Think about it like this. In New York City, there are just under four million residences and 74 of them catch fire every day. In your body, there are 30 trillion cells. If your cells are anything like New Yorkers, you are putting out 55 billion fires a day without even noticing it.

If a city suddenly decided to never put out fires, it would burn down to the ground tomorrow. Thus, while serotonin is usually absent from your mitochondria on the whole, it is always somewhere dealing with an emergency, and if you disrupt that process, your body will get into big trouble very quickly.

Serotonin Is Not a “Happy Chemical”

Serotonin is derived from the amino acid tryptophan and named after its appearance in serum (sero-) and its ability to regulate the tone of various body systems (-tonin). It was [first discovered](#) in 1935 when it was called “enteramine” due to its occurrence in the gut. It was named “serotonin” in 1948 by other researchers studying it in the blood, who found it in platelets and showed it to promote the constriction of blood vessels. In 1952, it was shown that “serotonin” and “enteramine” were the same chemical and the name “serotonin” eventually took over.

This name emphasizes its occurrence in the *blood*, not its occurrence in the *brain*.

The word “neurotransmitter” was not even used [until 1961](#). It was in the early 1950s that DW Woolley identified serotonin as an “antimetabolite” of LSD and carved out the role of serotonin as a major brain chemical. Woolley was a blind diabetic, and while there is [no evidence](#) that he took LSD himself, many have commented on the strength of his intuition and the great vision of his mind’s eye by referring to his ability to “see” — without working eyes — the “structure of serotonin in LSD.”

We now know that LSD, mescaline, and psilocybin all produce hallucinogenic effects at least in part by activating a subset of serotonin receptors known as 5-HT2A receptors.

In 1986, SSRIs were released for the treatment of depression, and the marketing of these drugs hijacked serotonin science.

The definition of serotonin as the “happy chemical” and a “mood booster” was pushed on a gullible public, but these terms have no basis in science.

Serotonin is best conceived as a **stress-coping chemical** that exerts its action primarily through **dissociation**. That is, it steps in to provide distance between the stressor and its target.

- When your food stresses out your gut, serotonin helps put distance between your gut and the food. At low levels, this speeds up your intestinal motility. At high levels, it causes vomiting and diarrhea.
- When you experience adverse consequences, serotonin helps separate you from those consequences by changing your future behavior. This is why [acute tryptophan depletion](#) abolishes the aversion to punishment in humans. The reason SSRIs can be helpful in [obsessive-compulsive disorder](#) is that OCD often involves a deficient response to punishment, so that when the brain is “punished” by the adverse consequences of the obsessions and compulsions it is unable to course-correct. But *excessive* aversion to punishment caused by excess serotonin would lead to excessive conflict avoidance and cowardice.
- When life stresses out your brain, serotonin helps put distance between you and your stress. When this is deficient, you over-identify with your stress and become overwhelmed. At the Goldilocks amount, you can separate your identity from your failures and misfortunes enough to live another day. At an excessive amount, it may be that you separate yourself from reality itself in a psychotic break. This is vaguely supported by research indicating that the hallucinogen-activated 5-HT2A receptors start decreasing in density prior to the [onset of schizophrenia](#), which might reflect a negative feedback response to overstimulation.
- A school shooter on SSRIs abhors his reality and dissociates himself from it by destroying it.

The fact that acute tryptophan depletion does decrease [aversion to punishment](#) in healthy men and women but [doesn't hurt mood](#) in healthy men indicates serotonin's reputation as a mood-booster is at minimum so grossly oversimplified as to be very misleading. However, tryptophan depletion does hurt mood [in women](#) and in men at risk for depression. Thus, it would be wrong to deny that serotonin plays a role in mood altogether.

Serotonin in fact [plays diverse roles](#) in the brain including aggression, anxiety, addiction, appetite, memory, mood, pain perception, sleep, body temperature regulation, cognition, imagination, learning, perception, fear, the gag reflex, nausea, and vomiting.

SSRIs have made the research base irrationally biased toward the brain. 95% of serotonin is in the gut, which is related to its role in the gut, not its role in the brain. While there is crosstalk between the brain and the gut, gut serotonin exists first and foremost to regulate gastrointestinal function, not mood. Its second role is to become a reservoir for serotonin to the entire body outside of the brain, providing the circulation with serotonin that is stored in platelets until it is needed. The ultimate fate of this serotonin is to be broken down in the liver or the lungs.

If you just look at where [serotonin transporters](#) are found, they are not remotely isolated to neurons nor are they found primarily or even disproportionately in the brain. They are most abundant in enterocytes (the primary cells of the small intestine) and apart from this they are widely distributed among neurons, skin cells, glandular secretory cells, muscle cells of the heart and blood vessels, ciliated cells of the lungs, immune cells, and others. **Serotonin is fundamentally a whole-body chemical, not a brain chemical.**

Serotonin Helps Your Mitochondria Cope With Hypoxic Stress

The primary function of serotonin in energy metabolism is to **help mitochondria dissociate from hypoxia by providing an alternative to oxygen during hypoxic stress.**

You *always* have some degree of hypoxic stress occurring in your body. Many people are anemic and therefore subject to chronic hypoxia. Some of your tissues will always be a little less perfused than they should be, because you are sitting the wrong way or you are engaged in a task that is so hungry for oxygen some tissues are not getting their needs met. You may be engaged in exercise that is raising the demand for oxygen far above rest. You may have sleep apnea. Chances are you are not breathing as well as you could be. Even if you are, every breath you take will oxygenate some sections of your lungs better than others, and your lungs are always trying to match blood flow to the most oxygenated area so that oxygen delivery remains efficient. *All* of these things involve some activation of the hypoxia response.

You take a breath. Parts of your lungs are very well oxygenated but other parts aren't. Serotonin kicks into gear shutting down blood flow to the poorly oxygenated parts. It *dissociates* the hypoxia in lung segments from hypoxia in blood by diverting blood flow to the most oxygenated parts of the lungs.

But when hypoxia reaches the mitochondrial level, this is when serotonin *really* steps up its game. And here, it is performing a remarkable feat, partly on its own, and partly by alerting melatonin to action, that preserves the function of the mitochondrial respiratory chain — where you make most of your cellular energy (ATP) — in the absence of oxygen.

Serotonin is doing this by 1) activating mitochondrial serotonin receptors and 2) entering the mitochondrion and becoming melatonin which then 3) exits the mitochondrion and acts on mitochondrial melatonin receptors and 4) inside the mitochondrion does something else mysterious.

And for this reason I hereby dub serotonin the **firefighter of the mitochondria** because it comes only in an emergency, whereas melatonin is the **mitochondria's guardian angel** because it is always waiting within the mitochondrion in watch of such an emergency.

The usual reaction to oxygen deprivation is to shut down respiration and switch on anaerobic glycolysis, oxidizing glucose to lactate and no further. This is very inefficient but it allows you to make some ATP.

In general you *want* to make this switch during hypoxia, because trying to maintain respiratory chain activity is energy-intensive and if you don't have the fundamental ingredient to make it work — oxygen — it makes no sense to try.

And this is the mysterious part: serotonin and melatonin act on the mitochondria to make this seemingly futile feat — keeping the respiratory chain going with no oxygen — a success.

One thing is established: both serotonin and melatonin have to act on receptors on the outside of mitochondria to preserve respiration in the face of hypoxia.

The other remains a mystery: it makes no sense that these molecules would *regulate* the preservation of respiration during hypoxia unless they were also *enabling* the preservation of respiration during this condition. So they *have* to be doing something else in addition to this. Otherwise wasting energy trying to do something that doesn't work would be cellular suicide.

So what is it?

We still need to study this but there are two possibilities I see:

- Melatonin is oxidized to the melatonyl cation radical, which substitutes for oxygen by taking electrons from cytochrome C. This would allow us to obtain 80% of the usual amount of ATP, which is 13 times more than we get from anaerobic glycolysis.
- Melatonin or its oxidation products are shifting ultraviolet light produced during hypoxia to longer-wavelength light that enables complex IV to substitute nitrite for oxygen, preserving 100% of normal ATP production.

Neither of these have direct experimental support but they are both possible explanations.

In the next installment, we take a look at the details of the science on just how serotonin facilitates this protective effect on your mitochondria under conditions of stress. You can read the next installment [here](#):

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How Serotonin Helps You Breathe



We cannot understand what SSRIs are really doing without understanding how serotonin is impacting mitochondrial energy metabolism.

[Read full story](#) →

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Jason M 4d

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"When life stresses out your brain, serotonin helps put distance between you and your stress. When this is deficient, you over-identify with your stress and become overwhelmed. At the goldilox amount, you can separate your identity from your failures and misfortunes enough to live another day. At an excessive amount, it may be that you separate yourself from reality itself in a psychotic break. This is vaguely supported by research indicating that the hallucinogen-activated 5-HT2A receptors start decreasing in density prior to the onset of schizophrenia, which might reflect a negative feedback response to overstimulation."

This helps explain why child abuse and child neglect contribute to schizophrenia

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♥ Liked by Chris Masterjohn, PhD

🔥 Fascinating series so far, I'm really enjoying it! Thank you!

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